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MILITARY SPECIFICATION

PUNCH, CUTTING, REVOLVING HEAD

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

* 1.1 This specification covers a plier-like handtool with six, different size, nonremovable cutting tubes assembled on a revolving head for hand punching holes in leather.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-P-416 —Plating, Cadmium (Electrodeposited).
QQ-Z-325 —Zinc Coating, Electrodeposited, Requirements for.

MILITARY

MIL-H-15424 —Hand Tools; Packaging of.

STANDARDS

FEDERAL

Fed. Test Method Std. No. 151—Metals; Test Methods.

MILITARY

MIL-STD-105 —Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 —Marking for Shipment and Storage.
MIL-STD-130 —Identification Marking of U. S. Military Property.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 **Preproduction sample.** When specified (see 6.2), before production is commenced, a sample of the item covered by this specification shall be submitted or made available to the contracting officer or his authorized representative for approval in accordance with 4.2. The approval of preproduction sample authorizes the commencement of production but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification. The preproduction sample shall be manufactured in the same facilities to be used for the manufacture of the production items.

3.2 **Standard product.** Cutting punches furnished under this specification shall be a current product of manufacturer, except for modifications from the manufacturer's design as may be specified herein.

3.3 **Illustrations.** The figure shown herein is for illustration only and is not intended to exclude punches otherwise conforming to this specification.

3.4 **Material.** The materials shall be as specified herein. Material not definitely speci-

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fied shall be of a composition and quality suitable to meet the requirements specified herein.

3.4.1 *Handles.* Handles shall be of steel or malleable iron.

* 3.5 **Design and construction.** The punch shall consist of handles and jaws, a revolving head, index spring, anvil, and six, nonremovable cutting tubes (see figure 1). When assembled, the punch shall pass the proof load test specified in 4.4.1.

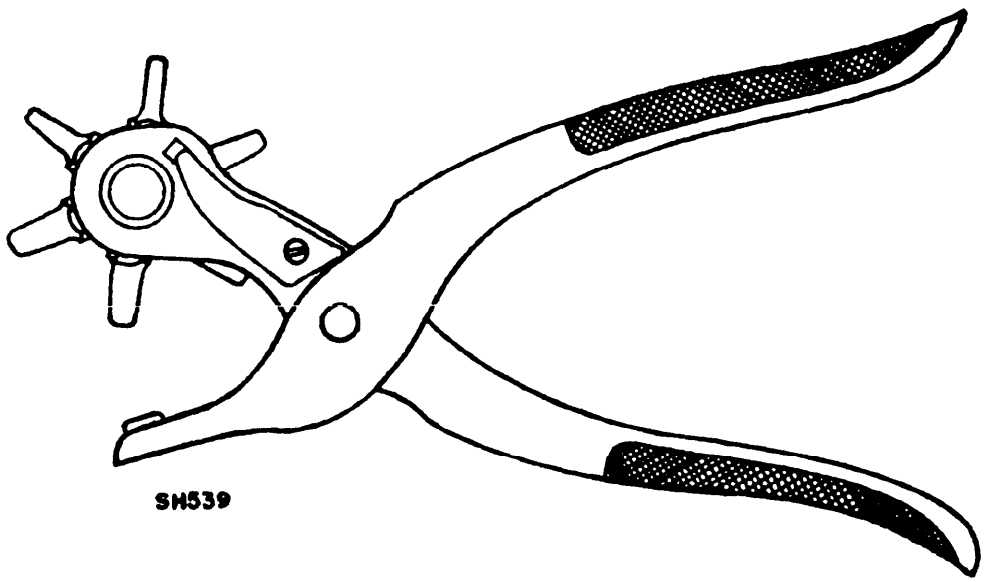


FIGURE 1. *Punch, cutting, revolving head.*

The punch shall be not less than 8 nor more than 9 inches long. The punch shall be designed to brace the revolving head firmly in the upper jaw and the anvil in the lower jaw. When stamped steel handles are supplied, the grip portion of the handles shall be knurled to prevent slippage. A spring made of spring steel shall be fitted to the handles in a manner that will cause the jaws to remain in an opened position when not under pressure. A stop shall be provided to preclude the handles spreading beyond 5 inches when measured 3¾ inches from the fulcrum rivet.

* 3.5.1 *Index spring.* An index spring of spring steel shall be securely fastened to the upper jaw of the handles. The index spring shall be tensioned sufficiently to position the head at any of the index stations on the revolving head without play.

3.5.2 *Revolving head.* The revolving head shall be fabricated from steel, and shall have six index stations. The stations, when aligned by the index spring, shall hold the cutting tubes perpendicular to the center of the face of the anvil and central within 1/32 inch when the punch is in a closed position. The head shall have six through holes equally spaced around its periphery to accept the cutting tubes. The center of the head shall be hollow to allow for discharge of the cutting blanks.

3.5.3 *Anvil.* The anvil shall be of soft copper or brass. The anvil shall be properly proportioned with a head having a minimum thickness of 1/16 inch, and 3/8 inch ±1/16 inch in diameter. The anvil shall be securely attached to the lower jaw.

3.5.4 *Cutting tubes.* The cutting tubes shall be fabricated from tool steel and shall be tightly assembled into the head to prevent removal or loosening during normal usage. The cutting tubes shall be sharpened to a

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TABLE I. *Nonremovable, fixed cutting tubes, diameter**

First station	Second station	Third station	Fourth station	Fifth station	Sixth station
Inch 0.078	Inch 0.094	Inch 0.109	Inch 0.125	Inch 0.156	Inch 0.188

* Tolerance shall be ±0.005 inch except the grip portion of the handles.

cutting edge capable of cutting clean, smooth holes. The cutting tubes shall be through hardened or case hardened to a hardness of Rockwell C50 to 60. The interior surfaces of the cutting tube shall be smooth and free from burrs and pits to allow unobstructed passage of cutting blanks for discharge through the open end of the tube opposite the cutting edge. The inside diameters of the cutting tubes shall be as shown in table I.

3.6 **Finish.** The punch shall be smoothly finished, and shall have one of the coatings specified in 3.6.1 through 3.6.3 as specified (see 6.2). The coating shall be adherent, smooth, continuous, and free from uncoated areas, pits, blisters, and nodules.

* 3.6.1 *Cadmium.* Cadmium plate shall conform to type II, class of QQ–P–416.

* 3.6.2 *Zinc plate.* Zinc plate shall conform to type II, class 3 of QQ–Z–325.

* 3.6.3 *Hot phosphate.* The coating shall consist of a heavy phosphate treatment.

* 3.7 **Marking.** Each punch shall be marked in accordance with MIL–STD–130.

* 3.8 **Workmanship.** The finished punch shall conform to the quality and grade of product established by this specification. The punch shall be free from rust, burrs, and sharp edges (except cutting edges).

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 **Preproduction sample inspection.** When a preproduction sample is required it shall be examined for the defects in 4.3.3.1, dimensions specified and tested in accordance with 4.4.1.

* 4.3 **Sampling for inspection and acceptance.** Sampling for inspection and acceptance shall be made in accordance with the provisions set forth in MIL–STD–105, except as otherwise indicated hereinafter.

* 4.3.1 *Inspection of components and materials.* In accordance with 4.1 above, the supplier is responsible for insuring that components and materials used were manufactured, tested and inspected in accordance with the specified requirements of referenced subsidiary specifications and standards, or, if none, in accordance with this specification.

4.3.2 *Intermediate inspection.*

* 4.3.2.1 *Hardness.* Cutting tubes shall be tested for hardness as specified in Method No. 243.1 of Fed. Test Method Std. No. 151 to determine compliance with 3.5.4. At least two readings shall be taken from each sample unit and any reading falling outside the specified limits shall be classed a defect. A lot shall consist of all cutting tubes of one size offered for acceptance at one time. The sample unit shall be one cutting tube. The inspection level shall be S–2 with an acceptable quality level (AQL) of 1.5 defects, expressed in terms of defects per hundred units.

* 4.3.3 *End product inspection.* The lot shall be the number of cutting punches offered for

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acceptance at one time. The sample unit shall be one complete cutting punch.

* 4.3.3.1 *Visual examination.* The examination of the end item shall be in accordance with the classification of defects and AQL's set forth below. The inspection level shall be level II with an AQL of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

Examine	Defect	Classification	
		Major	Minor
Finish	Missing	X	
	Blistered, peeled, chipped, scratched or foreign matter imbedded		X
	Indication of rust	X	
Design	Any characteristic not in accordance with specified requirements	X	
Construction and workmanship, (general).	Part missing	X	
	Sharp burr or edges (except cutting edge)	X	
	Fractured, split, sprung, cracked, malformed, dented, bowed, broken or otherwise impaired.	X	
	Component parts not connected or joined as specified	X	
	Forgings not free from scale, inclusions, cold shuts or seams (when applicable).		X
Marking identification	Missing, incomplete, not legible		X

4.3.3.2 *Dimensional examination.* Examination shall be made of the end item to determine compliance with dimensional requirements. Any dimension that fails to meet the specified requirements shall be classified a defect. The inspection level shall be S–2 with an AQL of 4.0 defects, expressed in terms of defects per hundred units.

4.3.3.3 *End item testing.* Test shall be performed on each sample punch as specified in 4.4.1. Any evidence of failure shall be classified a defect. The inspection level shall be S–2 with an AQL of 1.5 defects, expressed in terms of defects per hundred units.

4.3.4 *Examination of preparation for delivery.* An examination shall be made to determine compliance with preservation, packaging, packing and marking requirements in accordance with section 5 and MIL–H–15424.

4.4 Tests.

* 4.4.1 *Proof load test.* The punch shall be in a closed position with the cutting edge of the cutting tube resting against the face of the anvil. A force of 125 pounds shall be exerted on the handle, perpendicular to the vertical axis of the fulcrum, 3¾ inches from the fulcrum rivet. The decrease of the handle spread shall not be more than ¼ inch. After the force is released the handle shall show no permanent set.

5. PREPARATION FOR DELIVERY

* 5.1 Preservation, packaging and packing for level A, B, or C shall be as specified in MIL–H–15424 (see 6.2).

* 5.2 Marking. In addition to any special marking required by the contract or order, unit packages, intermediate packages, and shipping containers shall be marked in accordance with MIL–STD–129.

6. NOTES

6.1 Intended use. The punch covered by this specification is intended for use in leatherworking shops to punch holes in leather.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

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(b) When preproduction sample is required (see 3.1).

(c) Type coating required (see 3.6).

(d) Selection of the applicable levels of preservation and packaging and packing (see 5.1).

6.3 **Notations.** The outside margins of this document have been marked to indicate where changes from the previous issue have been made. An asterisk (*) indicates a substantive change in the paragraph text. A double asterisk (**) indicates deletion of a paragraph bearing the same number. The sign (#) indicates the addition of a completely new paragraph. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content as written irrespective of the marginal notations and relationship to the last previous issue.

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